

Lab 1 – Requirements Table: PES2UG24CS493 – Freelance Content Creator Escrow Platform

Domain: Media, Events & Community | Target Actors: Content Creator, Client, Sponsor

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SRN: PES2UG24CS493

SEC: H

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a Client to create an escrow contract for a project, specifying scope, milestones, and payment amount.	High	When a Client submits contract details with at least one milestone and an amount greater than 0, the system creates the contract and displays a unique Contract ID within 3 seconds.	Establishes the core agreement that anchors the escrow relationship between the Client and Content Creator.	<i>Clarify what happens if scope or milestones are edited after the contract is created — is versioning or Client/Creator re-approval needed?</i>
FR-002	Functional	The system shall allow a Client to deposit funds into escrow for an active contract using a supported payment method.	High	After successful payment authorization, the contract's escrow balance updates to reflect the deposited amount and a "Funded" status is shown within 5 seconds.	Funds must be secured in escrow before creative work begins, protecting both Client and Content Creator.	<i>Specify which payment methods are supported (card, UPI, wallet) and how partial or multiple deposits toward the same milestone are handled.</i>
FR-003	Functional	The system shall allow a Content Creator to submit a deliverable (file or link) against a specific contract milestone.	High	Upon submission, the milestone status changes to "Submitted - Pending Review" and the Client receives a notification within 1 minute.	Enables creators to formally hand off completed work for evaluation and payment release.	<i>Add constraints on deliverable submission — max file size, allowed file types, or accepted link domains.</i>
FR-004	Functional	The system shall allow a Client to approve or reject a submitted deliverable within a configurable review period (default 5 business days).	High	A Client action of "Approve" or "Reject" updates the milestone status accordingly and is time-stamped in the audit log within the review period.	Gives Clients formal control over deliverable quality while creating an auditable review trail.	<i>Define the system's behavior if the Client neither approves nor rejects within the review period — does it auto-approve or escalate?</i>

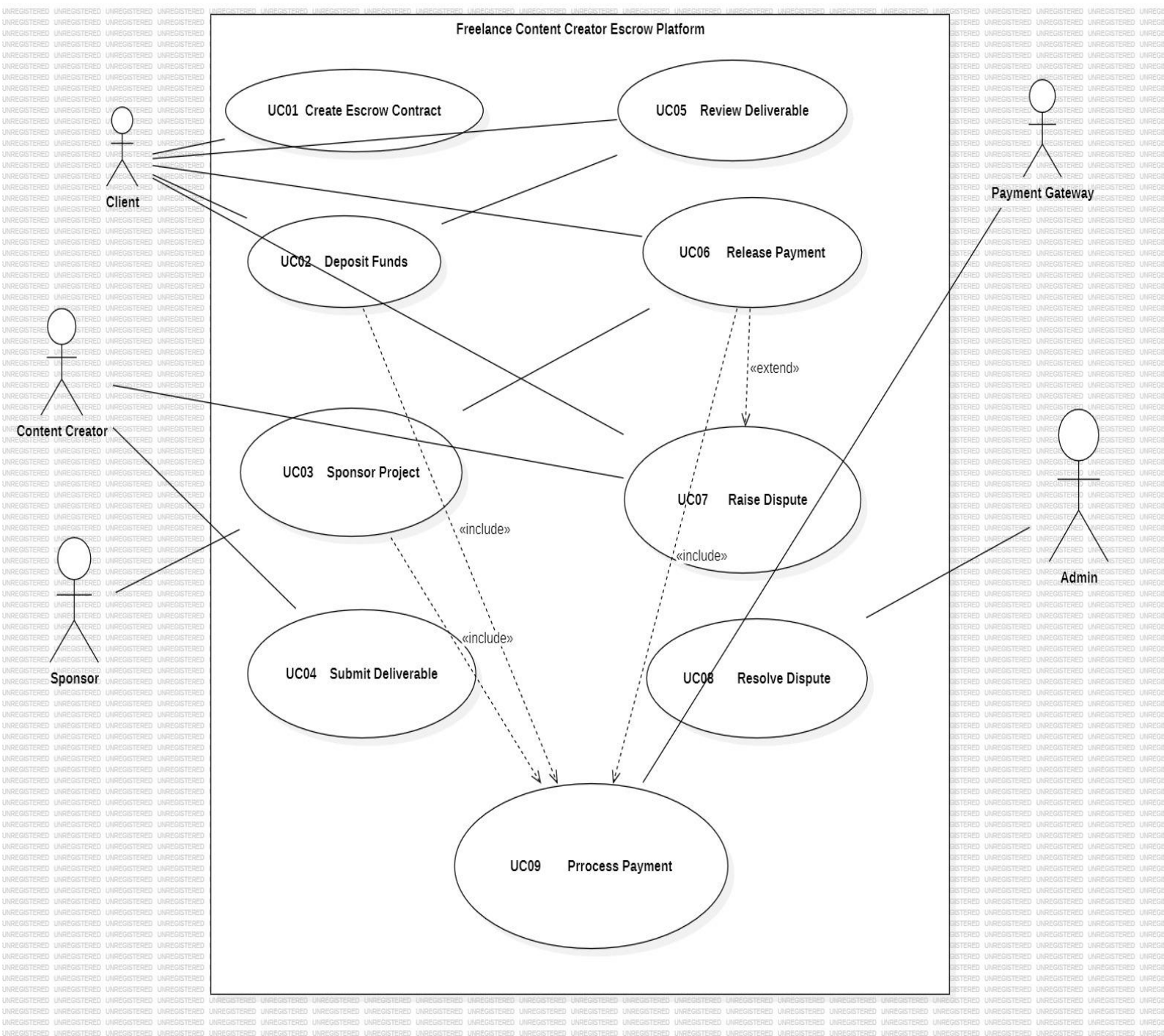
FR-005	Functional	The system shall allow a Sponsor to cofund a Client's project by contributing a specified amount to the project's escrow balance.	Medium	When a Sponsor confirms a contribution, the project's total escrow balance increases by the contributed amount and the Sponsor's name appears in the project's funding list within 5 seconds.	Supports community/sponsorbacked content creation, a key differentiator of the platform.	<i>Clarify whether a Sponsor's identity is always public in the funding list or can be contributed anonymously.</i>
NFR-001	Nonfunctional	The system shall process any escrow deposit, release, or withdrawal transaction within 5 seconds under normal load (up to 500 concurrent users).	High	Load testing with 500 concurrent simulated transactions shows a 95thpercentile transaction time of less than 5 seconds.	Ensures a responsive experience for time-sensitive financial actions during peak usage.	<i>Confirm the 5-second target also holds for write-heavy actions like payment release, not just deposits, since release involves an external gateway call.</i>
NFR-002	Nonfunctional	The system shall encrypt all financial and personal data, both in transit and at rest, using industry-standard encryption (TLS 1.2+ and AES-256).	High	A security scan confirms all API endpoints enforce TLS 1.2+ and database inspection confirms AES-256 encryption at rest for all PII and payment fields.	Protects sensitive financial and personal data, essential for user trust and regulatory compliance.	<i>Specify the key rotation policy and whether PCI-DSS compliance is required given third-party payment gateway integration.</i>

USE CASE FLOW DIAGRAM

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Lab 1 – Use-Case Flow Document

Field	Value
Name	Shreyash M Yadwad
SRN	PES2UG24CS493
Section	H
Project Title	Freelance Content Creator Escrow Platform

Use-Case Flow Specification

Goal	Release Escrowed Payment to Content Creator
Actor	Client (primary); Payment Gateway, Content Creator (supporting)
Main flow	1. Client navigates to an approved milestone and selects the "Release Payment" option. 2. The system displays the milestone amount, platform fee, and net payout to the Content Creator. 3. The Client confirms the release request. 4. The system validates that the milestone status is "Approved" and that sufficient escrow balance exists. 5. The system invokes Process Payment to transfer funds from escrow to the Content Creator's linked payout account via the Payment Gateway. 6. The Payment Gateway authorizes and processes the transfer. 7. The system updates the milestone status to "Paid" and decrements the escrow balance accordingly. 8. The system generates a payment receipt and notifies both the Client and the Content Creator. 9. The release of payment is complete.
Alternate Flow	Payment Gateway Declines Transfer At Step 6, if the Payment Gateway declines the transfer: 6a. The system displays a "Payment Transfer Failed" message to the Client. 6b. The Client is prompted to retry the release or update the Content Creator's payout details. 6c. If the transfer fails again after 2 attempts, the release is cancelled, the milestone status reverts to "Approved," and both parties are notified. Client Raises a Dispute Before Release At Step 1, if the Client raises a dispute instead of confirming release: 1a. The system places the milestone on hold and changes its status to "Disputed." 1b. The Admin is notified to review and resolve the dispute. 1c. The Client cannot release payment until the dispute is resolved.